

# Vansh Raja

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## ACADEMIC QUALIFICATION

**BTech CSE with spec. in Artificial Intelligence, Vellore Institute of Technology - Bhopal;**

Grade: 8.89; Aug 2025

## EXPERIENCE

**Security Analyst, Virtual Galaxy Infotech Ltd.**

**[Feb '26 – Current]**

*Mumbai, India (Remote)*

- Built Vaani, VGIL's multilingual voice agent (**lead developer**): seven Indian languages (Hindi, English, Marathi, Gujarati, Malayalam, Bengali, Odia) with mid-call switching, real inbound and outbound phone calls (SIP), and a dashboard so tenants can design and run their own agents instead of waiting on engineering (LiveKit, Sarvam STT/TTS, Gemini).
- Built a live RAG POC for Konkan Railway (**lead developer**): 20 operating and engineering manuals (~6,500 pages), so staff can ask in plain language and get the rule plus the exact PDF page, instead of hunting through the books.
- Built a live POC for Sir J.J. Hospital, Byculla (**lead developer**): a radiology asset register with AMC and warranty tracking, and a neurology MRI desk that takes OPD and IPD bookings off paper day-sheets.
- Worked with the infrastructure division on business development with other clients.

**Tools Used:** LiveKit, Sarvam AI, Gemini, Python, FastAPI, React, SIP, PostgreSQL, PGVector, Next.js, Docker

**Security Analyst Intern, Virtual Galaxy Infotech Ltd.**

**[Oct '25 – Jan '26]**

*Mumbai, India (Remote)*

- Built a RAG legal drafting system: ingested 5,000+ judgments into PGVector, ran PDF → metadata → chunking → embeddings, and served hybrid retrieval (vector + PostgreSQL FTS) with petition-aware drafting via Streamlit.
- Developed Windows/Linux DLP agents to monitor directories, clipboard, and PII; forwarded events in real time to a FastAPI backend with PostgreSQL, Redis, and OpenSearch for incident logging and policy actions.

**Tools Used:** RAG, PGVector, PostgreSQL, FTS, Python, FastAPI, Streamlit, Redis, OpenSearch, Windows, Linux, DLP, PII Monitoring

**Data Analyst Intern, PreProd Corp**

**[Nov '23 – Jan '25]**

*Bengaluru, India (Remote)*

- Built an end-to-end MLOps pipeline for food demand forecasting (Kafka, DVC, MLflow, PySpark, Scikit-Learn).
- Delivered demo projects for the Opensrc program using PySpark, RAG, GAN, MobileNet, Redis, and CouchDB.
- Ran a 2-day buildathon at VIT Bhopal; built the AutoML MVP, mentored teams, and reviewed submissions.
- Led 20 interns across 6 teams to ship 30+ demos; acted as Scrum Master and kept sprints on track.

**Tools Used:** Kafka, DVC, MLflow, PySpark, Scikit-Learn, Pandas, Polars, RAG, GAN, MobileNet, Redis, CouchDB, Computer Vision, AutoML

## PROJECTS

**KRCL Manual RAG**

**[Jun '26 – Aug '26]**

Live POC for Konkan Railway. Indexes 20 railway manuals (~6,500 pages) into a citation-first RAG chatbot so operating and engineering staff can look up rules, paras, and forms without flipping PDFs. Hybrid search with page-level citations.

**Tools Used:** Python, FastAPI, PostgreSQL, PGVector, RAG, React

**HMS JJH**

**[Jun '26 – Aug '26]**

Live POC for Sir J.J. Hospital, Byculla. Radiology asset management (inventory, AMC/warranty, service alerts) and a neurology MRI appointment system for OPD and IPD — kiosk, desk, and queue display instead of paper registers.

**Tools Used:** Next.js, TypeScript, PostgreSQL, React, Docker

**Vaani**

**[Jan '26 – May '26]**

Real-time multilingual voice agent for Virtual Galaxy. Speaks Hindi, English, Marathi, Gujarati, Malayalam, Bengali, and Odia, and can switch language mid-call. Production phone calling (SIP), visual agent builder, and a multi-tenant admin UI, on LiveKit with custom Sarvam STT/TTS and Gemini.

**Tools Used:** Python, LiveKit, Sarvam AI, Gemini, FastAPI, React, SIP, PostgreSQL

## **Data Loss Prevention** [<https://github.com/effaaykhan/Data-Loss-Prevention>]

[Nov '25]

Proof-of-concept DLP system with cross-platform agents (Windows/Linux) that monitor files, clipboard, and USB devices. Detects sensitive data like Aadhaar numbers, API keys, and passwords. Features a React dashboard for live events, agent status, and policy management. Includes Google Drive integration.

**Tools Used:** Python, FastAPI, PostgreSQL, Redis, OpenSearch, Docker, Windows, Linux, React

## **CERTIFICATIONS**

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- “Privacy and Security in Online Social Media” by NPTEL, Apr '24
- “AI and ML” by Google Developers & SmartInternz, Nov '23
- “Applied Machine Learning in Python” by University of Michigan, Feb '23

## **TECHNICAL SKILLS**

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**AI & ML:** Scikit-learn, Polars, PySpark, Dask, H2O AutoML, PyCaret, TPOT, Langchain, Ollama, Matplotlib, Seaborn, Plotly, LLM APIs (Groq, OpenAI, Google AI), LiveKit, SIP, RAG, Prompt Engineering

**Frontend:** Tailwind CSS, Framer Motion, Vite, Radix UI, Streamlit, Dash

**Databases:** Hadoop, Kafka, SQL, PostgreSQL, PGVector, Cassandra, MongoDB, Redis, Neo4j, CouchDB

**DevOps & Other:** Flask, FastAPI, MLflow, Docker, Oracle Cloud, Git, Scrum